

## Section 0 – Title and contact Information

- **Title of the proposed solution track:** Precision nutrient management at scale: Supporting Vietnam's One-Million-Ha high-quality and low-emission rice program
- **Title of the AoW2 Flagship that the solution track aligns with:** Flagship 2.1 - Scaling Seed and Soil Fertility Management Systems for Resilient, Diverse, and Nutrition-Smart Farming
- **Lead Scientist(s) / Contact Person(s) / Center(s):**

Dr. Hung Van Nguyen, Senior Scientist, focal person of IRRI for the One-Million-Ha program, IRRI

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Dr. Anton Urfels, Senior Scientist, Water Management, IRRI

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- Key CGIAR Center Partners (in the case of multi-Center Eols): International Potato Center (CIP)
- **Technical link to other Science Programs:** Sustainable Farming (SFP)

## Section 1 – Innovation Package, Value Proposition & Alignment

**Purpose:** To address the problems of rice production in Vietnam, using excessive nitrogen driving high input cost, low income, and high carbon footprint and water pollution, this project will scale precision nutrient management (PNM)—building on site-specific nutrient management (SSNM), bundled mechanized direct seeding combined with fertilizer application (mDSR), novel fertilizers, bio-inoculants, and straw recycling, that enhance the performance of the core innovation and support the national program on low-emission rice. By improving profitability, reducing emissions, and maintaining yield quality, PNM directly benefits smallholder farmers, particularly women and disadvantaged groups who bear disproportionate labor burdens and are vulnerable to fluctuations in input prices. Anchored in Vietnam’s One-Million-Ha high-quality, low-emission rice program, this scaling solution will deliver both national and global impact. By 2028 (Year 3 of this project), we aim to reach 600,000 ha and benefit over 600,000 farmers, ultimately reaching 1M ha and over 1M farmers by 2030.

- **The problem to be solved:** Describe the challenge you are seeking to overcome through the proposed Scaling Solution track. What country / countries will the Solution Track operate in? Be sure to clarify who you are seeking to benefit with this work, with attention paid to gender and social inclusion. (~150 words)

Vietnam produces about 44 million tonnes of rice, exporting 6-7 million tonnes annually and contributing significantly to global food security. Yet, rice production faces persistent challenges: fertilizer use is often inefficient, with high input costs<sup>1</sup> eroding farmer incomes, while excessive nitrogen application drives greenhouse gas emissions and water pollution. The government initiatives, such as the “One Must Do, Five Reductions (1M5R),” have shown strong policy momentum toward low-input, high-quality production, but adoption is constrained by the lack of scalable, precision-based technologies<sup>2</sup>. This project will scale precision nutrient management (PNM)—building on site-specific nutrient management (SSNM)<sup>3</sup>. Bundled with this innovation are specific practices such as mechanized direct seeding combined with fertilizer application (mDSR)<sup>4</sup>, novel fertilizers, bio-inoculants, and straw recycling<sup>5</sup>, that enhance the performance of the core innovation and support the national program on low-emission rice. By improving profitability, reducing emissions, and maintaining yield quality, PNM directly benefits smallholder farmers, particularly women and disadvantaged groups who bear disproportionate labor burdens and are vulnerable to input price fluctuations. Anchored in Vietnam’s One-Million-Ha high-quality, low-emission rice program, this scaling solution will deliver both national and global impact<sup>6</sup>.

- **Your scaling ambition:** Describe the ambition of your proposed Scaling Solution track. Which of the KPIs do you believe you will contribute most towards, and at what level of quantitative and qualitative impact? (~150 words)

The scaling ambition is to transform rice production in Vietnam’s Mekong River Delta by embedding precision nutrient management (PNM) within the One-Million-Ha high-quality, low-emission rice program. Success will be measured through four KPIs. **Scale:** by 2028 (Year 3 of this project), we aim to reach 600,000 ha and benefit over 600,000 farmers, ultimately reaching 1M ha and over 1M farmers by 2030. **Productivity and Income:** farmers will achieve at least a 5% increase in net income through reduced fertilizer costs by at least 5% while maintaining yield. **Environmental impact:** PNM will reduce the carbon footprint of rice

cultivation by 10% (LCA approach) while maintaining grain quality and export competitiveness. **Inclusion:** gender equity and youth participation will be promoted through women-led business models, digital engagement, and learning alliances. Adoption will be driven by bundling mDSR, SSNM, novel fertilizers, bio-inoculants, and rice straw recycling. Supporting digital tools will further enhance extension services, accelerate adoption, and ensure inclusive, evidence-based decision-making at scale.

- **Innovation & Value Proposition** – Describe the innovation or bundle or package (maturity, novelty, suitability for scaling) and the benefits it delivers to institutional partners/customers and end-users (e.g., productivity, resilience, nutrition, risk reduction, institutional capacity). Can include linkages to other science programs that will collaborate and work on the innovation/bundle/package (~150 words)

This package builds on the proven principles of SSNM, already embedded in Vietnam's government recommendations (e.g. 1M5R), and advances toward Precision Nutrient Management (PNM) for large-scale transformation<sup>7</sup>. PNM enhances nutrient efficiency by integrating high-resolution soil nutrient mapping, digital analytics, and mechanized application. Mechanized direct seeding with precision deep fertilizer placement, deployed on ~1,000 ha and supported by technical guidebooks, increases nitrogen efficiency, profitability, and reduces emissions<sup>3,8,9</sup>. Novel fertilizers and bio-inoculants for rapid rice straw decomposition, already being validated, strengthen nutrient recycling, soil health, and emission mitigation<sup>11</sup>. Complementary AI-powered platforms—Lua Viet, EasyFarm, and MRV-FarMore—provide real-time advisory and monitoring<sup>8</sup>. This bundle is a core component of the CGIAR Sustainable Farming Systems Science Program in Southeast Asia. It delivers productivity, resilience, and environmental benefits to farmers while enabling institutional partners to scale evidence-based practices through Vietnam's One-Million-Ha high-quality, low-emission rice program.

- **Customers & Alignment** – Identify the main institutional customers (ministries, donor programs, agribusinesses, lenders) and end-users (farmers, SMEs, cooperatives). Show how the track aligns to an AoW2 Flagship and Grand Challenge. (~100 words)

The main institutional customers include the Ministry of Agriculture and Environment (MAE) and its related departments such as Plant Production and Protection (PPPD), National Agricultural Extension Center (NAEC), Vietnam Rice Sector Association (VIETRISA), Fertilizer Association (VFA), Farmer Union (VFU), Women Union (VWU), and related agribusinesses. End-users are primarily rice farmers, cooperatives, and stakeholders engaged in production and service delivery. This project aligns with the programs, including [World-Bank One-Million-Ha initiative](#), [USDA-Fertilize-Right](#), and [Binh Dien Fertilizer Straw-Soil](#) projects. The innovation bundle—centered on precision nutrient management and scaled through policy engagement, farmer capacity development, and business model innovations—directly aligns with AoW2 Flagship 2.1 (Seed and Fertilizer) under Grand Challenge 2, while fostering collaboration across CGIAR science programs and bilateral projects. The innovation bundle—centered on precision nutrient management and scaled through policy engagement, farmer capacity development, and business model innovations—directly aligns with AoW2 Flagship 2.1 (Seed and Fertilizer) under Grand Challenge 2, while fostering collaboration across CGIAR science programs and bilateral projects.

- **Evidence of Impact** – Provide citations and evidence of both effectiveness (impact) and scalability in practice (uptake beyond pilots, adoption drivers). (~150 words)

The innovation bundle has demonstrated strong evidence of impact during its pilot phase across 1,000 hectares in 2024–2025. The mDSR combined with fertilizer deep placement demonstrated its benefits of reducing 30-50% seeding rate, 20-30% N fertilizer, and 10-30% pesticide in rice production<sup>12</sup>; rice straw–based nutrient recycling added about 10% income into the rice value chain<sup>11</sup>. The principles of SSNM, already validated in Vietnam and incorporated into national recommendations, significantly improved yield and nitrogen use efficiency<sup>13</sup>. Its digital tailoring through the Rice Crop Manager (RCM) platform has enabled user-friendly, field-specific recommendations, reducing complexity and driving adoption at scale. With consecutive positive results from the field pilots under the One-Million-Hectare program, the PNM innovation built a solid foundation for the scaling phase<sup>14,15</sup>. By 2028, we aim to reach 600,000 ha and benefit over 600,000 farmers, ultimately reaching 1M ha and over 1M farmers by 2030. The project would substantially contribute to Vietnam’s climate and green growth targets<sup>9,10</sup>. It also serves as a replicable model for driving sustainable agricultural transformation across the region.

## Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

**Purpose:** To assess whether there is a credible pathway for the innovation to scale through public, private, or PPP delivery models — supported by sustainability plans, partner demand and commitments, financial leverage, and risk mitigation.

- **Scaling Pathway & Sustainability** – Describe how the innovation bundle and/or package will scale (public, private, PPP). Explain how the solution will remain viable at scale (revenues, public financing, PPP, IFI/donor support), with milestones or cost-recovery targets. (~263/250 words)

The PNM bundle will be scaled with the One-Million-Ha program through a PPP model, leveraging strong government commitment, donor support, and private sector engagement. MAE will lead policy alignment and public financing for implementing the program. Agribusinesses and cooperatives will drive adoption and service delivery. The multistakeholder linkage and PPP will be facilitated by the Vietnam Rice Sector Association with the support of IRRI and the National Agricultural Extension Center. The WB-Vietnam One-Million-Hectare program provides a foundational platform for scaling, complemented by the bilateral donor support (e.g., USDA, Mekong Cooperation Fund, Taiwan ICDF, etc.) and CGIAR’s science programs. Milestones:

- 2025-2026: Mapping scale-appropriate technologies, creating stakeholder engagement network and PPP, and developing site-specific strategy, workplan, and pathway for large-scale scaling based on the pilot results and lessons learned.
- 2026-2027: Scaling the innovation bundle in at least 200,000 ha.
- 2027-2030: Full-scale deployment to 1 million ha with integrated precision nutrient management, including mDSR, rice straw-based circular economy, digital tools, and branding for Vietnam green and low-emission rice.

Sustainability will be achieved through multiple financing streams. Farmers will benefit from revenue gains (≥5% income increase) through yield improvements, lower input costs, and value creation from rice straw recycling. Cost recovery will come from reduced seed, fertilizer, and

pesticide use; premium pricing for branded low-emission rice; and private sector investment in mechanization and nutrient services. Public financing will support capacity building, infrastructure, and climate-smart subsidies, while donor and S4I support will enable continued innovation and impact monitoring. Women-led business models, learning alliances, and digital platforms will foster inclusive entrepreneurship and farmer engagement, ensuring long-term viability, institutional ownership, and transformative impact in Vietnam's rice sector.

- **Partnerships & Leverage** – Identify internal CGIAR champions and external partners that have a stake in championing the innovation bundle or package, provide evidence of demand (surveys, requests for collaboration, alignment with policies or strategies) and commitment (MOUs, LOIs, budget allocations from partners, or other evidence of demand), and specify major non-CGIAR programs or investments this track can leverage (budgets, partners, policies, programs, etc.). (~300 words)

The innovation bundle is championed internally by IRRI scientists working across agronomy, soils and nutrient management, data science, innovation systems, and MELIA, including the focal point of the One-Million-Ha program. CIP works in the same intervention areas to promote rice-sweet potato crop rotation as Mekong is the largest sweet potato production region in Vietnam, bringing synergies for emission reduction, soil health and pest and disease mitigation. These CGIAR champions provide deep technical expertise, policy engagement, and institutional commitment to scaling.

Externally, MAE is a key partner, providing policy alignment and public financing. The Vietnam Rice Sector Association (VIETRISA) and the National Agricultural Extension Center (NAEC) are actively engaged in facilitating multi-stakeholder partnerships and PPP models. Agribusinesses, cooperatives, and farmer organizations such as the Farmer Union and Women's Union have expressed strong interest in adopting and promoting the innovation bundle, particularly through women-led business models and inclusive learning alliances.

Evidence of demand includes: (i) strong farmer and cooperative interest through pilot field demonstrations; (ii) requests from exporters for premium, green, and low-emission rice; (iii) formal collaboration requests from MARD and provincial governments; and (iv) alignment with national strategies such as Vietnam's Green Growth Strategy and the One-Million-Ha program<sup>2,12,14,15</sup>.

The innovation track leverages major non-CGIAR programs and investments, including the World Bank's investment in Vietnam's rice transformation, USDA, MKCF, TaiwanICDF, and Binh Dien Fertilizer, totaling about 7 million USD in 2026-2028.

For implementing the project, IRRI has MoUs with the key government partners and institutions, such as MAE, NAEC, Nong Lam University (NLU), Vietnam Academy of Agricultural Sciences (VAAS), etc. About 30% of the budget will be allocated to the partners through LOAs, including 6 Departments of Agriculture and Environment (DAE), VIETRISA, and related stakeholders to scale the One-Million-Ha innovation bundle. These partnerships and financial commitments provide a robust foundation for scaling, sustainability, and institutional ownership, ensuring transformative impact across Vietnam's rice sector.



- **Risks & Bottlenecks** – Identify specific risks (financial, political, operational, social, or others) and bottlenecks (policy, institutions, finance, technology) in the specific context and place that could hinder scaling. Describe potential mitigation strategies to address these risks and bottlenecks. (~250 words)

Scaling agricultural innovations in Vietnam faces risks and bottlenecks across financial, political, operational, institutional, technological, and social dimensions. Financially, limited access to affordable credit and investment hampers the adoption of mDSR, digital tools, and other innovations. Politically, shifts in government priorities or delays in policy enforcement can stall progress. Operational challenges include fragmented extension systems and uneven technical capacity, which reduce the effectiveness of knowledge transfer and farmer support.

Institutionally, weak coordination among ministries and slow regulatory processes hinder the scaling progress. Technological bottlenecks such as low digital literacy, poor rural connectivity, and mismatched technologies further constrain scaling efforts. Socially, resistance to change, gender disparities, and limited youth engagement pose additional barriers.

To mitigate these challenges, a blended financing model supported by the One-Million-Hectare program and private sector investment can unlock resources. Strong alignment with national strategies and continuous policy dialogue will ensure political support. Capacity building through farmer learning alliances and localized technical support, including digital platforms like E-extension Lua-Viet integrated AI-chatbot and PNM-RCM, can address operational gaps. Institutional coordination and streamlined policy frameworks will accelerate PPP-based scaling. Technological adoption can be enhanced through comprehensive scaling approaches and platforms like EasyFarm, which connect farmers with stakeholders. Finally, inclusive models—such as women-led businesses and youth-focused learning alliances—can foster broader social engagement and sustainability.

### Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

**Purpose:** To define what can realistically be achieved with S4I support in the next 2–3 years — including achievable reach, milestones, resources, activities, and safeguards for responsible scaling.

- **Readiness & Responsible Scaling** – Provide evidence of scaling readiness (IPSR or similar evidence). Describe actions for ensuring that scaling will be inclusive (extending to women, youth, marginalized groups) and environmentally sustainable. (~150 words)

The innovation bundle has demonstrated strong readiness for scaling. SSNM is a legacy of IRRI research and development over 40 years<sup>13</sup> (but still needs an integrated PNM bundled by novel fertilizers, mDSR, and digital tools, and enabling the One-Million-Ha program for large-scale scaling under S4I). RCM-Vietnam, mDSR, and rice straw-based circular economy have been piloted successfully across 1,000 hectares in 2024–2025 with measurable impacts on productivity, emissions reduction, and farmer income<sup>14,15</sup>. The technologies are mature, field-validated, and supported by institutional partners and national programs, including the World Bank-funded One-Million-Hectare initiative and the USDA-Fertilize Right projects. The scaling readiness is detailed in the innovation systems analysis for RCMAS<sup>16</sup> and RCM scaling strategy<sup>17</sup>.

Responsible scaling will be ensured through inclusive models such as women-led agribusinesses, inclusive learning alliances, and participatory extension platforms, building on the gender-responsive scaling approaches developed in 2024<sup>18</sup>. Through this project, we enable responsible scaling within the 1M Ha program and support activities that embed this principle into the systems driving innovation uptake. Environmental sustainability is embedded through climate-smart practices, circular economy models for rice straw, and integrated digital tools to track the project KPIs. This inclusive approach ensures that scaling is not only feasible but also socially and environmentally responsible.

- **Achievable target, Resources & Activities** – Define what can realistically be achieved in 2–3 years starting 2026 forward with S4I support and current/planned partnerships. Outline the key resources (staff, institutional, financial, technical) and activities (e.g., training, policy engagement, distribution) required. (~250 words)

This project, proposed in 3 years (2026-2028), supported by S4I aims to scale the Efficient Fertilization innovation bundle to support the One-Million-Hectare program to at least 1 million ha and benefit at least 1 million farmers across the Mekong Delta. Major activities, milestones, deliverables, and resources are shown in Table 1.

Table 1. Achievable target, resources & activities

| # | Activities                                                                       | Targeted Deliverables                                                                                                                                                                                                                                                   | Resources                                                                        |
|---|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1 | Mapping scale-appropriate technologies for scaling in MRD                        | Ecological- and value chain-based suitability mapping of the Efficient Fertilization technologies bundle                                                                                                                                                                | IRRI, MAE, and local research and development institutions                       |
| 2 | Developing site-specific strategy, workplan, and pathway for large-scale scaling | Integrated strategy and methodology for large-scale deployment of the Efficient Fertilization bundle based on the pilot results and lessons learned                                                                                                                     | IRRI, MAE, DAEs, and local research and development institutions                 |
| 3 | Field demonstrations, ToTs                                                       | Demonstration reports and implementation guidelines showcasing the combined package: mDSR + deep placement, upgraded digital nutrient tools, novel fertilizers/bioinoculants, rice-straw circular solutions, with national technical guidebooks and extension curricula | IRRI, MAE, DAEs, NAEC, VIETRISA, and local research and development institutions |
| 4 | Establish inclusive business models and learning alliances                       | Women-led enterprises and youth engagement networks operational involved in One-Million-Ha program                                                                                                                                                                      | Women Union, Farmer Union, IRRI innovation team                                  |

|   |                                                                         |                                                                                              |                                                                                                          |
|---|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 5 | Engage stakeholders for policy alignment and support                    | Policy frameworks and incentive mechanisms for Efficient Fertilization bundle implementation | MAE, provincial governments, Vietnam Rice Sector Association, IRRI                                       |
| 6 | Deploy and scaling Efficient Fertilization innovation bundle across MRD | Targeted 1 million ha in MRD                                                                 | One-Million-Ha technical guideline associated with mDSR, SSNM, AWD, rice straw management, digital tools |

- **Timeline, Milestones & Monitoring** – Provide an estimate of annual milestones for 2026–2028 (users reached, partnerships formed, policies adopted). Explain how progress will be monitored, lessons captured, and adaptations made. (~200 words)

In 2026, the focus will be on mapping scale-appropriate technologies, developing site-specific strategies, conducting field demonstrations, and capacity building - ToTs. In 2027, the innovation bundle will be deployed across 200,000 hectares, with policy frameworks adopted in key provinces. By 2028, the program will cover 600,000 hectares, ultimately reaching 1 million hectares and benefiting over 1 million farmers by 2030, with implementation of the core technologies in the PNM bundle, including SSNM, mDSR, and rice straw management.

A comprehensive monitoring and evaluation for the learning strategy will be established and implemented. The strategy will begin with a baseline study in 2026 to establish benchmark indicators, followed by an endline assessment in 2028, covering at least 5,000 farmer households through large-scale surveys.

Progress will be systematically tracked using key performance indicators (KPIs), including hectares brought under improved practices, number of users reached, reductions in greenhouse gas (GHG) emissions, and measurable improvements in farm household incomes. Consistent with the CGIAR Performance and Results Management System (PRMS) and other donor-reporting requirements, internal monitoring system (e.g. thru interactive templates) will be established to regularly track key deliverables to inform timely course-corrective actions.

In addition to quantitative tracking, lessons and insights will be continuously captured through field monitoring reports, learning alliances, and structured feedback loops with farmers and implementing partners. Cost-efficient qualitative outcomes assessments (e.g. case study, most significant change) will be conducted in 2027 to document emerging early outcomes.

The monitoring strategy will also be adaptive in nature, with findings feeding into iterative planning processes, policy dialogues, and technical refinements. This will ensure that scaling efforts remain evidence-based, responsive to farmer needs, and inclusive of diverse stakeholder perspectives.



**Annexes (See attached PDF file)**

- A. Scaling Readiness Assessment Results
- B. Innovation Packages and Scaling Readiness (IPSR)
- C. Business Model Canvas (BMC-lite)
- D. Prior Project Reports
- E. Media Coverage
- F. MELIA (Monitoring, Evaluation, Learning, Impact Assessment) Framework
- G. Gender and Inclusion
- H. F. References